

Study of Mathematical Modeling on Effect of Swine Flu

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Abstract: This mathematical model used Routh–Hurwitz Stability Criterion to provide the critical evaluation of the epidemics of Swine Flu. We introduce the Jacobian to determine the effect of variations in the potential of the epidemic for their prediction. Stability of infectious disease through this model is depend upon eigen value $\lambda_i >, =, < 0$; $i=1,2,3$ and found the given system marginally stable.

Keywords: Compartmental Model, SIR Model, Swine Flu.

I. INTRODUCTION

Swine Flu is highly contagious respiratory disease "refers to influenza caused by those strains of influenza virus, called the swine influenza virus, that generally infect pigs". Symptoms of swine flu are fever, which is usually high; cough; runny nose or stuffy nose; sore throat; body aches; headache; chills; fatigue or tiredness, which can be extreme; diarrhea and vomiting, sometimes, but more commonly seen than with seasonal flu. Signs of a more serious swine flu infection might include pneumonia and respiratory failure. By solving the governing equations of the model, we try to understand how changes in human behaviour and social mixing influenced the epidemic [1, 3, 13].

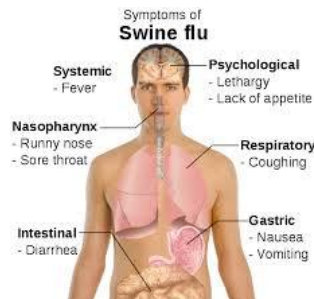


Fig. 1: Symptoms of Swine Flu

In the past few months, the number of diagnosed swine flu and influenza A (H1N1) cases has raised rapidly, in particular area of Himachal Pradesh. In this region, the number of confirmed infections has increased to 399 in the year 2013, but this only covers a small proportion of the cases which have occurred in the region [2]. The currently circulating strain of swine origin influenza virus of the H1N1 strain has undergone triple reassortment and contains genes from the avian, swine and human viruses. It is transmitted by droplets or fomites. Preventive measures include: social distancing, practicing respiratory etiquette, hand hygiene and use of chemoprophylaxis with antiviral drugs. Vaccine against H1N1 is not available at present, but will be available in near future [3]. The current swine flu caused by a novel H1N1 virus has infected a documented 1, 82,166 humans, killing 1,799 from April 2009 to August 2009. The World Health Organization (WHO) declared the outbreak to be a pandemic because of growing worldwide cases during the current H1N1 outbreak; infection control is recommended to prevent spread of the virus among humans. Public health authorities are developing action plans which may request social distancing actions depending on the severity of the outbreak. Mathematical models are often used to study disease spread, with the Susceptible- Infectious-Recovered (SIR) model being preferred for diseases spread via droplet and aerosol. The SIR model has been used to study pandemic flu, seasonal flu [4]. Many models for the

spread of infectious diseases in populations have been analyzed mathematically and applied to specific diseases. Threshold theorems involving the basic reproduction number R_0 , the contact number 0 , and the replacement number R are reviewed for the classic SIR epidemic and endemic models. Similar results with new expressions for R_0 are obtained for MSEIR and SEIR endemic models with either continuous age or age groups. Values of R_0 and a are estimated for various diseases including measles in Niger and pertussis in the United States. Previous models with age structure, heterogeneity, and spatial structure are surveyed [5, 12]. Massive mathematical models have been formulated and developed to study various infectious diseases. The history of epidemiology in ancient Indian literature such as the Rigveda (c. 8000 BC), Charaka Sahara (c. 700 BC) and several other Ayurvedic texts until 1600 AD, Puranas (c. 200 BC to 750 AD), travel accounts of visitors to India, and some British records. The compartment model SIR for dynamic models discussed by Kermack and Mckendrick in 1927 and are used by many other biomathematicians [6]. Swine flu is respiratory disease which is caused by type-A influenza virus of subtypes H1N1, H1N2, H3N1, H3N2 and H2N3 and discussed Kermack Mc Kendrick (transmission model) for the spread, effect and their optimal control rates [7, 8, 10, 12]. The A (H1N1) pdm09 seasonal virus is still active after the 2009 pandemic in Rajasthan [9]. The models can incorporate the birth, death, and immunization and analyze the outcome mathematically and SIR models including bifurcation analysis [11].

II. MATHEMATICAL MODELLING

The diagram of this compartment model is a system of differential equations which are designed for the susceptible to swine flu (S), infected with swine flu (I), Expose for swine flu (E) is given below:

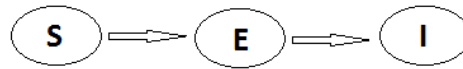


Fig. 2

Where, we consider that due to virus infection, the total number of susceptible decrease and number of infected person increases. We also assume that the transmission rate of swine flu of decrease S, or transmission rate of swine flu increase I, is the proportional to the product of the number of susceptible and the number of infected.

A. Mathematical Compartmental Model

Susceptible Class, S: Susceptible to swine flu, β contact rate of swine flu,

–

Expose Class, E: Exposed with swine flu, α rate of transmission due to swine flu,

–

Infective Class, I: Infected with swine flu, γ rate of recovery rate due to swine flu,

–

The total size of host population is $N = S + E + I$. Let us simplify the model which can be written as following;

— $(= \text{Say, } A) \dots\dots\dots (i)$

— $(= \text{Say, } B) \dots\dots\dots (ii)$

— $(= \text{Say, } C) \dots\dots\dots (iii)$

B. The Equilibrium Position of the Model

For equilibrium the system, we have

— — —

Consider the initial values of the parameters as $S = x$; $E = y$; $I = z$.

Then from the model, we get

$$\dots\dots\dots(iv)$$

$$\dots\dots\dots(v)$$

$$\dots\dots\dots(vi)$$

C. The Disease Free Equilibrium Point of the Model

Here, the disease free equilibrium point is that point where the infection is totally absent from the population and we have $z = 0$ from the equation (iv).

From (iv) we have $\Rightarrow$ — where $y = 0$ and $z = 0$.

Therefore the disease free equilibrium point (—) $\dots\dots\dots(vii)$

D. Mathematical Analysis on this Model

The Jacobian of this model is given by

$$\begin{matrix} - & & \dots\dots\dots(viii) \\ & - & \end{matrix}$$

The characteristics equation for the system of the governing equation is given by $|J - \lambda I| = 0$

$$\begin{matrix} - & & = 0 \\ & - & \end{matrix} \dots\dots\dots(ix)$$

By using the initial values from (vii) we have (—) = (—). Finally, we observed that

$$\begin{matrix} - & & \dots\dots\dots(x) \end{matrix}$$

Case 1: $\dots\dots\dots(xi)$

Case 2: $\dots\dots\dots(xii)$

Therefore the condition for stability of the model is that all the eigen values $\dots\dots\dots$; such that

If $\dots\dots\dots$, then the system is stable.

If $\dots\dots\dots$, then the system is unstable.

If $\dots\dots\dots$, then the system is stable.

E. Stability Analysis of the Model

We have the equation from for a third order polynomial. From the equation (ix), we have

;

Where ; ; .

Since all the coefficients are satisfied the Routh–Hurwitz Stability Criterion. We have the table which has all the first column elements are positive, zero and the same sign. Therefore, the given system is marginally stable. Because

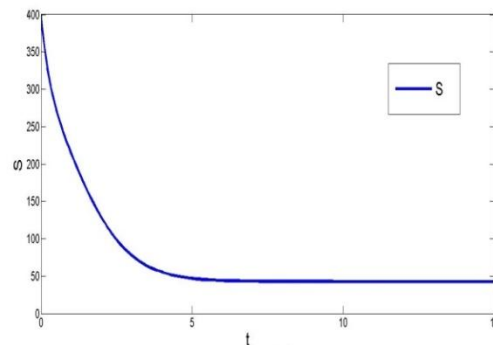
F. Numerical Result of the Model

Parameters	Values	Units (per/time)
α , transmission rate of swine flu	0.12531	Year
β , Contact rate of swine flu	0.05764	Year
, removal rate from the infection	2.46269	Year

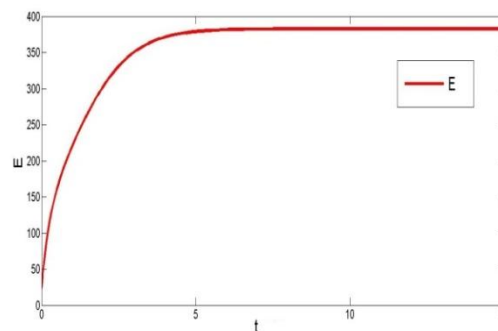
Table: 2013 Swine Flu parameters values of the model

Now using Routh's table with the initial values $S_0 = 399$, $E_0 = 23$, $I_0 = 23$, we got the zero's in the first column.

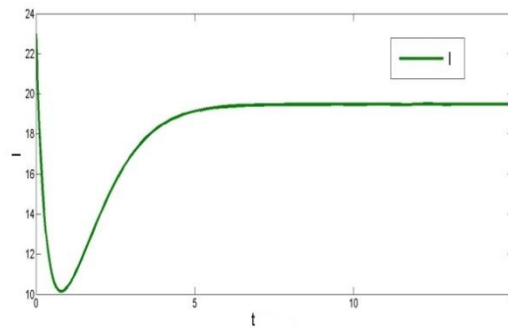
III. Graph



Graph.1



Graph. 2



Graph. 3

III. CONCLUSION

In this model, Matlab draw the figure 1, 2 and 3 for variables S, E, I respectively, which shows numbers of susceptibility is decreasing, numbers of exposed is increasing countionously while this model decreases I for a short time period and than increases with respect to time.

Finally, we have obtained the useful results on swine flu by using Routh–Hurwitz Stability Criterion, that the given model is marginally stable. The graphical output of the paper is similar to other standards results. Also, This paper is providing the future perdition and medication from H1N1 virus in the population to medical policy makers.

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